Computer Networks

Unit 1 – Introduction: Summarised Notes

1. Introduction to Computer Networks

Network

A network is a set of devices, often referred to as nodes, that are connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Computer Network

A computer network is defined as a set of computers interconnected by communication links for the purpose of sharing data, sharing resources, or exchanging messages. Communication here means sharing information, and this sharing may be local (face-to-face) or remote (taking place over a distance).

Data Communication

Telecommunication means communication at a distance. The term data refers to information presented in whatever form is agreed upon by the parties creating and using it. Data communication is therefore the exchange of data between two devices via some form of transmission medium, such as a wire cable.

Characteristics of Data Communication

- **Delivery:** When a sender transmits data, it must reach only the intended destination.
- **Accuracy:** The system must deliver the data accurately.
- **Timeliness:** The system must deliver data without any delay.
- **Jitter:** Jitter refers to the variation in packet arrival time.

Components of Data Communication

- **Message:** The information (data) to be communicated, e.g. text, numbers, pictures, audio, or video.
- **Sender:** The device that sends the data message, e.g. a computer, telephone handset, or video camera.
- **Receiver:** The device that receives the message, e.g. a computer, telephone handset, or television.
- **Transmission Medium:** The physical path by which a message travels from sender to receiver, e.g. twisted pair wire, coaxial cable, fibre optic cable, or radio waves.
- **Protocol:** A set of rules that governs data communications and represents an agreement between communicating devices; without it, two connected devices cannot understand each other.

Data Flow

Data flow describes the direction in which signals can travel between two connected devices. There are three modes of data flow.

- **Simplex:** Communication is unidirectional — only one device sends and the other only receives.
- **Half-duplex:** Each device can send and receive, but not at the same time.

- **Full-duplex:** Both devices can send and receive simultaneously.

Network Criteria

- **Performance:** Depends on the network elements and is measured in terms of delay and throughput.
- **Reliability:** Measured by the failure rate of network components, i.e. its availability and robustness.
- **Security:** Protection of data against corruption or loss, whether from errors or from malicious users.

Physical Structures

- **Type of Connection – Point to Point:** A dedicated link between exactly one transmitter and one receiver.
- **Type of Connection – Multipoint:** A single transmission is shared among multiple recipients.
- **Physical Topology:** Describes how devices are connected and the type of transmission used — unicast, multicast, or broadcast.

2. Networking Topologies

Physical Topology

Physical topology refers to the way in which a network is laid out physically. Two or more devices connect to a link, and two or more links form a topology. A topology is the geometric representation of the relationship of all the links and linking devices (nodes) to one another.

Mesh Topology

In a fully connected mesh topology, every node is connected to every other node. If there are n nodes, each node must connect to n – 1 other nodes, requiring a total of n(n – 1)/2 physical links (full duplex requires n(n-1) links). A practical example is the connection of telephone regional offices, where each regional office needs to be connected to every other regional office.

Advantages

- Eliminates traffic problems that occur with shared links.
- Robust — if one link becomes unusable, it does not disable the entire system.
- Provides privacy or security, since each message travels along a dedicated line.

Disadvantages

- Requires a large amount of cabling and many I/O ports.
- Installation and reconnection are difficult.
- Bulk wiring can exceed available space (in walls, ceilings, or floors).
- Hardware (I/O ports and cabling) can be prohibitively expensive.

Star Topology

In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. Devices are not directly linked to one another; all communication passes through the hub.

Advantages

- Less expensive than a mesh topology.
- Easy to install and reconfigure.
- Requires less cabling to be housed.
- Robust — if one link fails, only that link is affected while all others remain active.

Disadvantages

- The whole topology depends on a single point, the hub; if the hub goes down, the entire system fails.
- Mainly suited for use within local-area networks (LANs).

Bus Topology

A bus topology is a multipoint communication method in which one long cable acts as a backbone to link all devices in the network. Nodes are connected to the bus cable through drop lines and taps. A drop line runs between the device and the main cable, and a tap is a connector that splices into or punctures the cable to make contact with the metallic core. As a signal travels along the backbone, some of its energy converts to heat, so the signal weakens the farther it travels — limiting the number of taps and the distance between them.

Advantages

- Ease of installation.
- The backbone cable can be laid along the most efficient path and connected to nodes using drop lines of varying length.
- Uses less cabling than mesh or star topologies, since only the backbone runs through the entire facility.

Disadvantages

- Difficult reconnection and fault isolation.
- Difficult to add new devices once installed for optimal efficiency.
- Signal reflection at the taps can cause degradation in quality.
- Adding new devices may require modification or replacement of the backbone.

Ring Topology

In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it. A signal passes along the ring in one direction, from device to device, until it reaches its destination. Each device incorporates a repeater — when it receives a signal meant for another device, it regenerates the bits and passes them along.

- Relatively easy to install and reconfigure, since each device is linked only to its immediate neighbours.
- Adding or deleting a device requires changing only two connections.
- Constrained mainly by media and traffic considerations (maximum ring length and number of devices).
- Fault isolation is simplified: if a device does not receive an expected signal within a specified period, it can issue an alarm.

3. Categories of Networks

Local Area Network (LAN)

A local area network (LAN) is usually privately owned and links devices within a single office, building, or campus. A LAN can be as simple as two PCs and a printer in a home office. LAN size is currently limited to a few kilometres, and it typically uses only one type of transmission medium. The most common LAN topologies are bus, ring, and star. Early LANs had data rates of 4 to 16 Mbps, whereas speeds today are normally 100 or 1000 Mbps; wireless LANs are the newest evolution in LAN technology.

Wide Area Network (WAN)

A wide area network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may span a country, a continent, or the whole world. A

WAN can be as complex as the backbones connecting the Internet or as simple as a dial-up line connecting a home computer to the Internet.

- **Switched WAN:** Connects end systems, usually through a router (an internetworking device), to another LAN or WAN.
- **Point-to-Point WAN:** Normally a leased line from a telephone or cable TV provider connecting a home computer or small LAN to an Internet Service Provider (ISP); commonly used to provide Internet access.

Metropolitan Area Network (MAN)

A metropolitan area network (MAN) is a network whose size falls between a LAN and a WAN. It normally covers the area inside a town or a city and is designed for customers who need high-speed connectivity, usually to the Internet, with endpoints spread over a city or part of a city.

Internetwork

An internetwork is formed by the interconnection of networks — for example, a heterogeneous network made up of several WANs and LANs connected together to allow communication across otherwise separate networks.

4. Protocol and Network Architecture

Protocol

A protocol is synonymous with a rule. It consists of a set of rules that govern data communications, determining what is communicated, how it is communicated, and when it is communicated. The key elements of a protocol are syntax, semantics, and timing.

- **Syntax:** The structure or format of the data; indicates how to read the bits.
- **Semantics:** Interprets the meaning of the bits and identifies which fields define which action.
- **Timing:** Specifies when data should be sent and at what speed it should be sent or received.

Network Architecture

A computer network must provide general, cost-effective, and robust connectivity among a large number of computers. To manage this complexity, network designers developed the concept of network architecture, which introduces layered protocols and functions that guide the design and implementation of networks. The most widely used network architectures are the OSI Architecture and the Internet Architecture.

5. The OSI Reference Model

OSI Model

The Open Systems Interconnection (OSI) model is an ISO standard covering all aspects of network communications. It was first introduced in the late 1970s. Note that ISO is the organization, while OSI is the model itself. The OSI model organizes communication into seven layers, split into three subgroups: Layers 1–3 (physical, data link, network) are the network support layers, dealing with the physical aspects of moving data between devices; Layers 5–7 (session, presentation, application) are the user support layers, implemented in software, which allow interoperability among unrelated software systems; and Layer 4 (transport) links the two subgroups and ensures end-to-end reliable data transmission.

Principles Used to Arrive at the Seven Layers

- A layer should be created where a different level of abstraction is needed.
- Each layer should perform a well-defined function.
- The function of each layer should be chosen with a view toward defining internationally standardized protocols.
- Layer boundaries should be chosen to minimize the information flow across the interfaces.
- The number of layers should be large enough that distinct functions are not forced together unnecessarily, yet small enough that the architecture does not become unwieldy.

Physical Layer

The physical layer is responsible for the movement of individual bits from one hop (node) to the next. It coordinates the functions required to carry a bit stream over a physical medium, dealing with the mechanical and electrical specifications of the interface and transmission medium, and defining the procedures and functions that physical devices and interfaces must perform for transmission to occur.

- **Physical characteristics of interfaces and medium:** Defines the characteristics of the interface between devices and the transmission medium, and the type of medium used.
- **Representation of bits:** Data is a stream of bits (0s and 1s) with no inherent meaning; to be transmitted, bits must be encoded into electrical or optical signals, and the physical layer defines the type of encoding used.
- **Data rate:** Defines the transmission rate — the number of bits sent each second.
- **Synchronization of bits:** The sender and receiver must use the same bit rate and have synchronized clocks.
- **Line configuration:** Concerned with the connection of devices to the media, either point-to-point (dedicated link between two devices) or multipoint (a link shared among several devices).
- **Physical topology:** Defines how devices are connected to form a network — mesh, star, ring, bus, or hybrid topology.
- **Transmission mode:** Defines the direction of transmission between two devices — simplex, half-duplex, or full-duplex.

Data Link Layer

The data link layer is responsible for moving frames from one hop (node) to the next, making the physical layer appear error-free to the network layer above it. It is responsible for node-to-node delivery — transmitting a group of bits, called a frame, to the adjacent node without error. To send data across several hops, the frame is delivered in stages between successive pairs of adjacent nodes, with headers and trailers changing at each hop.

- **Framing:** Divides the stream of bits received from the network layer into manageable data units called frames.
- **Physical addressing:** Adds a header to the frame to define the sender and receiver when frames must be distributed to different systems on the network.
- **Flow control:** Imposes a mechanism to prevent the sender from overwhelming the receiver when the receiver absorbs data more slowly than the sender produces it.
- **Error control:** Detects and retransmits damaged or lost frames and prevents duplication, achieved through a trailer added at the end of the frame.
- **Access control:** Determines which device has control over the shared link at any given time when two or more devices are connected to the same link.

Network Layer

The network layer is responsible for the delivery of individual packets from the source host to the destination host. If two systems are connected to the same link, a network layer is usually unnecessary; but if the systems are attached to different networks with connecting devices between them, the network layer is needed to accomplish source-to-destination delivery.

- **Logical addressing:** When a packet passes a network boundary, an additional addressing system for the source and destination — the logical address — is required.
- **Routing:** Devices called routers connect various independent networks and are responsible for delivering packets to their final destination, using routing tables to decide the next hop.

Transport Layer

The transport layer is responsible for process-to-process delivery of the entire message. A process is an application program running on a host. Unlike the network layer, which treats each packet independently, the transport layer ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

- **Port addressing:** The header includes a port address (service-point address) so the message reaches the correct process on the destination computer.
- **Segmentation and reassembly:** The message is divided into segments, each assigned a sequence number, allowing the receiving side to reassemble them in the correct order.
- **Connection control:** Can be connectionless — where each segment is treated as an individual packet — or connection-oriented, where a connection is established before delivery and terminated afterwards.
- **Flow and error control:** Similar to the data link layer, but performed on a process-to-process basis rather than node-to-node.

Session Layer

The session layer is the network's dialog controller. It establishes, maintains, and synchronizes the interaction among communicating systems — that is, it establishes, manages, and terminates connections between applications. This is needed because the services provided by the physical, data link, and network layers alone are not sufficient for some processes.

- **Dialog control:** Allows two systems to enter into a dialog, either in half-duplex or full-duplex mode.
- **Synchronization:** Allows checkpoints to be added into a stream of data — for example, inserting checkpoints every 100 pages within a 2000-page file transfer so that if a crash occurs, only the pages since the last checkpoint need to be resent.

Presentation Layer

The presentation layer is responsible for translation, compression, and encryption. It is concerned with the syntax and semantics of the information exchanged between two systems.

- **Translation:** Converts messages between the different encoding systems used by different computers into a common format, enabling interoperability.
- **Encryption and decryption:** The sender transforms the original information into another form before sending it over the network, and the process is reversed at the receiving end.
- **Compression and expansion:** Reduces the number of bits contained in information, particularly text, audio, and video.

Application Layer

The application layer enables the user, whether human or software, to access the network. It is responsible for providing services to the user, offering interfaces and support for services such as

electronic mail, remote file access and transfer, and shared database management, along with other distributed information services (e.g. X.400 message handling service, X.500 directory services, and FTAM for file transfer, access, and management).

- **FTAM (File Transfer, Access, and Management):** Allows a user to access files on a remote host.
- **Mail services:** Provide email forwarding and storage.
- **Directory services:** Provide database sources to access information about various network sources and objects.